
Beta Testing Document

for

IITK ProjectSpace

Version 1.0

Prepared by

Group 4:

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
1.0	All team members	Revised the entire document for clarity, consistency, and structure. Improved language flow, corrected grammatical issues, and ensured alignment with the intended purpose and audience. All sections have been reviewed and updated where necessary.	17/04/26

1 Introduction

This document outlines the comprehensive beta-testing evaluation conducted by our team for **IITK ProjectSpace**, a web-based software application developed by Team 3 (Kernel Panic). As the designated beta-testing group for this phase of the software development lifecycle, our objective was to rigorously evaluate the platform's functionality, usability, and reliability to ensure it meets its stated goals.

From our perspective as students and evaluators within the IIT Kanpur ecosystem, we recognize the critical campus problems this software aims to address. Currently, the campus lacks a unified platform for project management and academic discovery. Innovative projects developed by the IITK community often remain hidden in isolated GitHub repositories, stored locally on personal devices, or confined to specific club rooms, resulting in little to no visibility. Furthermore, the process of recruiting students for academic or extracurricular projects is highly fragmented. Professors and student organizations currently rely heavily on stray Google Forms, scattered personal websites, or inefficient cold-mailing to find collaborators. This extreme decentralization also leads to a significant loss of academic knowledge and institutional memory whenever contributing students graduate.

The primary purpose of IITK ProjectSpace is to eliminate these bottlenecks by serving as a self-contained, centralized hub for project showcasing, discovery, and recruitment within the academic community. By consolidating these disparate tools into a single environment, the application aims to connect undergraduate, postgraduate, and PhD students, alongside researchers, clubs, and faculty members. The platform's strategic goal is to foster a culture of open collaboration while preserving the community's technical work by archiving it in a searchable and verifiable repository.

To fulfill this purpose, Team 3 implemented a range of specific functionalities that served as the primary targets for our systematic testing phase. The platform relies on a secure authentication workflow restricted to the campus community via IITK webmail OTP verification. Once authenticated, users interact with a dynamically updated feed of public projects and open recruitments.

Team 3's Software Requirements Specification (SRS) document explicitly identifies the beta testing team as part of its intended audience. Therefore, guided by their specifications, our team simulated real-world usage scenarios. This report details the issues identified during our Sub-Phase I exploratory testing, the subsequent fixes implemented by the development team, and our final evaluation of the software's readiness to serve the institute.

2 List of Reported Bugs

BUG 1:

Tested Feature: Eligibility Enforcement

Tester Name: Krishna Agrawal

Testing Date: 13 April 2026

Bug Details:

- **Test Case:** Attempt to submit an application for a Recruitment Post when the user does not meet the "Eligibility" criteria defined in the post.
 - **Steps to Reproduce:**
 1. Log in to the IITK ProjectSpace platform.
 2. Navigate to the **Recruitments Page** (as described in Section 8.1 of the User Manual).
 3. Identify a recruitment post where the "Eligibility" field (e.g., "Y22 only" or "Proficiency in React") does not match the logged-in user's profile.
 4. Click the **Apply** button.

Expected Result: As the User Manual implies that only "eligible users" should apply (Section 8.2), the system should perform a server-side check. The backend should compare the **User** model attributes (batch, skills) against the **Recruitment** model's eligibility strings and return a **403 Forbidden** or a validation error if they do not match.

Actual Result: The system allows the application to proceed. The frontend triggers the "Application Submitted" success toast, and a new entry is created in the database. No automated validation is performed by the FastAPI backend to enforce the requirements stated in the recruitment post.

Bug Report Date: 9 April 2026

Has the bug been fixed? Yes

Date of Bug fixing: 10 April 2026

Comment: The fix is satisfactory. The developers have now implemented a validation layer in the recruitment application endpoint. This aligns the software with the "Future Development Plans" mentioned in the Implementation Document regarding making the recruitment process smoother for the recruiter by reducing ineligible applications.

BUG 2:

Tested Feature: Third-Party Authentication (Sign in with Google)

Tester Name: Ayush Mangal

Testing Date: 13 April 2026

Bug Details:

- **Test Case:** Attempt to authenticate using the "Sign in with Google" button on the login page.
 - **Steps to Reproduce:**
 1. Navigate to the Login page.
 2. Click on the "Sign in with Google" button.
 3. Select an active @iitk.ac.in Google account from the pop-up/redirect.

Expected Result: The system should verify the Google OAuth token, match it with an existing user (or create a new profile), and redirect the user to the dashboard.

Actual Result: The "Sign in with Google" button is unresponsive. The implementation document lists Google Auth in the tech stack, but the integration appears incomplete or the OAuth client credentials are not properly configured in the environment variables.

Bug Report Date: 10th April

Has the bug been fixed? Yes

Date of Bug fixing: 12th April

Comment : This is a critical UX issue as many students prefer SSO over manual OTP-based registration. The backend needs to properly register the callback URL in the Google Cloud Console and ensure the FastAPI endpoint for Google callback is active.

BUG 3:

Tested Feature: User Authentication / Login & Registration Forms

Tester Name: Rushali Vijayakumar Myageri

Testing Date: 9th April

Bug Details:

- **Test Case:** Toggle password visibility during entry in Login/Signup forms.
 - **Steps to Reproduce:**
 1. Navigate to the Registration or Login page.
 2. Enter a string into the "Password" field.
 3. Look for a "visibility" or "eye" icon within or near the password input field to reveal the characters.

Expected Result: The UI should provide a toggle button (typically an eye icon) that allows the user to switch the input type from "password" to "text" to verify the characters they have typed, preventing typos.

Actual Result: There is no option to reveal the password. The characters remain obscured as dots/asterisks at all times. This is particularly problematic for the "Set Password" field during registration, where a typo could lead to the user being locked out immediately after account creation.

Bug Report Date: 9th April

Has the bug been fixed? Yes

Date of Bug fixing: 10th April

Comment : While not a functional backend failure, this is a significant usability issue for a project space used by many students. Implementing a simple state toggle in the React/Next.js frontend for the input type would greatly improve the user experience.

BUG 4:

Tested Feature: Login Authentication System

Tester Name: Aman Gupta

Testing Date: 13 April 2026

Bug Details:

- **Test Case:** Attempt to log in using only the username.
 - **Steps to Reproduce:**
 1. Open the login page.
 2. Enter only the username (without `@iitk.ac.in`).
 3. Enter the password.
 4. Click **Login**.

Expected Result: Users should be able to log in using only the username as mentioned in the instructions.

Actual Result: Login fails when only the username is entered.

The system requires the full email format (`username@iitk.ac.in`) instead of just the username

Bug Report Date: 9 April 2026

Has the bug been fixed? Yes

Date of Bug fixing: 10 April 2026

Comment: The issue has been resolved in the latest commit. After the fix, users are able to log in using only their username as intended. The bug fix is satisfactory.

BUG 5:

Tested Feature: Website Performance / System Response Time

Tester Name: Abubakar Siddique

Testing Date: 14 April 2026

Bug Details:

- **Test Case:** Check responsiveness of the system when clicking buttons or navigating the website.
 - **Steps to Reproduce:**
 1. Open the website.
 2. Click on any button or interactive element (e.g., navigation buttons or login).
 3. Observe the system response time.

Expected Result: The system should respond immediately or within a short acceptable delay after a user action.

Actual Result: The website takes a significant amount of time to respond after clicking. This causes unpredictable behaviour and makes the UI appear slow and glitchy.

Bug Report Date: 10 April 2026

Has the bug been fixed? Yes

Date of Bug fixing: 10 April 2026

Comment: The issue related to system slowness has been resolved. After the fix, the website responds faster and the UI behaviour is stable. The fix significantly improves the overall user experience.

BUG 6:

Tested Feature: User Profile / Data Entry Form Validation

Tester Name: Ankit Kumar Jha

Testing Date: 14 April 2026

Bug Details:

- **Test Case:** Check whether **Designation, Degree, and Department** are required during user registration and before requesting a project.
 - **Steps to Reproduce:**
 1. Register as a new user on the platform.
 2. Complete the registration.
 3. Attempt to request a project without filling these profile details.

Expected Result: The system should require users to fill **Designation, Degree, and Department** either during registration or before allowing them to request a project.

Actual Result: The system allows users to register without entering Designation, Degree, or Department. Previously, users were also able to request projects without completing these details, which made it difficult to determine whether the user was eligible for the project.

Bug Report Date: 10 April 2026

Has the bug been fixed? Yes

Date of Bug fixing: 10 April 2026

Comment: The issue has been partially resolved. Users are now required to complete their profile details before requesting a project. However, the system still allows registration without filling these fields, which may lead to incomplete user profiles.

BUG 7:

Tested Feature: Testing FAQs

Tester Name: Aditi

Testing Date: 15th April 2026

Bug Details:

- **Test Case:** Check for FAQ availability on the project and recruitment pages.
 - **Steps to Reproduce:**
 1. Navigate to the Create section.
 2. Locate the instruction line at the top of the page.
 3. Attempt to find or click on the FAQs mentioned in the instruction.

Expected Result: The FAQs should be visible or accessible as prompted by the top instruction line.

Actual Result: The top line instructs the user to read FAQs, but the FAQs are missing or have not been implemented yet.

Bug Report Date: 7 April 2026

Has the bug been fixed? Yes

Date of Bug fixing: 12 April 2026

Comment: The issue has been fixed successfully. The FAQs are now available and accessible as indicated by the instruction on the page.

BUG 8:

Tested Feature: Dashboard Interactivity

Tester Name: Sahil Agarwal

Testing Date: 15th April 2026

Bug Details:

- **Test Case 1:** Verify the functionality of the comment button on the dashboard.

- **Steps to Reproduce:**

1. Navigate to the main dashboard.
2. Locate any post or section with a comment button.
3. Click the comment button.

Expected Result: The comment section should open, or a text input field should appear.

Actual Result: The comment button does not respond or trigger any action.

- **Test Case 2:** Verify the redirection of user profiles on the dashboard.

- **Steps to Reproduce:**

1. Navigate to the dashboard.
2. Click on a specific user's name or profile icon on any article or post .

Expected Result: The system should redirect to the selected user's profile page.

Actual Result: Clicking on a user leads the system to open the related article instead of the profile.

Bug Report Date: 8th April 2026

Has the bug been fixed? Yes

Date of Bug fixing: 10th April 2026

Comment: The dashboard navigation and button responsiveness have been corrected. Both the comment triggers and user profile links are now functioning as expected.

BUG 9:

Tested Feature: Profile Name Validation

Tester Name: Nideesh H

Testing Date: 15th April 2026

Bug Details:

- **Test Case:** Check for character restrictions in the Full Name field.
 - **Steps to Reproduce:**
 1. Go to the profile section.
 2. Click on "Edit Profile."
 3. Enter "Full Names : -1" (or any string containing numbers and special characters) into the name field.
 4. Save the changes.

Expected Result: The system should prevent the use of numbers and certain special characters to maintain readability and consistent referencing.

Actual Result: Usernames can be changed to anything (e.g., "-1"), making them unreadable and difficult to reference within the system.

Bug Report Date: 10th April 2026

Has the bug been fixed? Yes

Date of Bug fixing: 10th April 2026

Any other comment (for example, how satisfied are you with the bug fixing): The fix is effective. Names are now restricted to letters, spaces, dots, hyphens, and apostrophes, which ensures professional formatting and better readability.

3 Overall Quality of the Software

Overall Satisfaction:

Overall, the software provides a useful platform for managing and showcasing projects within the IITK community. The system successfully centralizes project discovery, recruitment, and collaboration into a single interface, reducing reliance on scattered tools such as Google Forms or personal repositories. The platform is easy to navigate and provides most of the core features required for project management and recruitment.

1. How good is the user manual? Was it easy to install and run the software?

The user manual is well structured and clearly explains the features of the system, including registration, login, search functionality, project pages, recruitment pages, and profile management. It provides step-by-step instructions for performing tasks such as creating posts, applying to recruitments, and editing user profiles. This makes it easy for new users to understand how to use the platform.

Since the software is a web-based application, installation is minimal for end users. Users simply need a browser and internet connection to access the platform, which makes the system easy to run and accessible across multiple devices.

2. How is the quality of the code? Is it well documented? Is the code modular? Are the variables and function names meaningful?

Based on the documentation and system structure, the software appears to follow a modular design with clear separation between frontend and backend components. The system uses modern technologies such as ReactJS for the frontend, FastAPI for the backend, and PostgreSQL for data storage, which promotes maintainability and scalability.

The architecture suggests that different modules handle specific tasks such as authentication, profile management, search functionality, project management, and recruitment management. This modular design helps improve readability, maintainability, and extensibility of the system.

3. Do you think that the software lacks some major features that may make it less acceptable to the user?

While the platform provides most of the essential features for project management and recruitment, there are some features that could further improve the user experience. For example:

- Stronger validation of user profile information during registration
- Improved filtering and recommendation systems for projects and recruitments
- Real-time messaging or collaboration tools between users
- More advanced analytics or tracking for project engagement

Adding such features could make the platform more interactive and useful for the IITK community.

4. Does the software not satisfy any major non-functional requirements?

The software satisfies most of the core functional requirements described in the documentation, including authentication, search, recruitment management, and project posting. However, during testing, some performance issues such as slow response times were observed when interacting with certain elements of the interface.

Improving performance optimization and response time would help ensure a smoother user experience and better adherence to non-functional requirements such as efficiency and responsiveness.

Appendix A - Group Log

Date	Timings	Duration	Minutes
4th April 2026	03:00 pm to 4:00 pm	1 hour	Finalized the User Manual. Transmitted the System Requirements Specification (SRS), Implementation Document, and User Manual to Team 5 for Sub-Phase I beta testing.
5th April 2026	10:00 am to 12:30 pm	2 hours 30 minutes	Reviewed and analyzed the SRS, Implementation Document, and User Manual provided by Team 3 to establish testing parameters.
6th April 2026	08:00pm to 11:00 pm	3 hours	Conducted systematic testing of Team 3's software based on their documentation; initiated identification and logging of functional bugs and UI/UX issues.
7th April 2026	09:00 pm to 11:00 pm	2 hours	Continued rigorous exploratory beta testing of Team 3's application; reproduced and documented edge-case anomalies.
8th April 2026	03:00 pm to 04:30 pm	1 hour 30 minutes	Performed further system testing on Team 3's software and drafted detailed descriptions and reproduction steps for the identified issues.
9th April 2026	02:30 pm to 04:30 pm	2 hours	Concluded Sub-Phase I testing. Finalized and officially submitted comprehensive bug

			reports via Team 3's GitHub repository issue tracker
10th April 2026	03:00 pm to 05:00 pm	2 hours	Transitioned to Sub-Phase II: Reviewed and triaged the bug reports submitted by Team 5 regarding TrackMate. Delegated bug-fixing tasks.
11th April 2026	9:00 pm to 10:45 pm	1 hour 45 minutes	Executed backend and frontend code revisions to resolve issues reported by Team 5; conducted local testing on the implemented fixes.
12th April 2026	01:00 pm to 7:00 pm	6 hours	Finalized all bug fixes ahead of the Sub-Phase II deadline. Pushed stable commits to the repository and closed resolved issues on the tracker.
13th April 2026	8:00 pm to 11:00 pm	3 hours	Initiated compilation of the final Beta Testing Report. Tracked and verified the status of the bug fixes implemented by Team 3 on their repository.
15th April 2026	9:00 pm to 02:00 am	5 hours	Drafted comprehensive sections of the final report, detailing testing methodologies, issue resolution status, and overall evaluation of Team 3's software.
16th April 2026	9:00 pm to 12:00 am	3 hours	Finalized formatting and proofread the Beta Testing Report; prepared the final deliverable to ensure submission prior to the April 17 deadline.

